



Process orientation: conceptualization and measurement

Process
orientation

267

Markus Kohlbacher and Stefan Gruenwald
Campus 02 University of Applied Sciences, Graz, Austria

Abstract

Purpose – Attention to processes has increased, as thousands of firms have adopted the process view of their organization. Process orientation (PO) means focusing on business processes rather than emphasizing functional structure or hierarchy. Despite the huge growth in the business process management literature, a methodological gap still remains about a certain ambiguity in the definition of the construct specifying its theoretical domain and dimensionalization. The aim of this paper is to contribute to the literature on PO through the creation and validation of a model to measure the key dimensions of the PO construct.

Design/methodology/approach – The authors derive from literature several dimensions which shape the concept of PO. In a second step, the study includes factor analysis on a sample of 152 Austrian manufacturing companies to verify that the model is empirical valid.

Findings – The paper provides insight in the concept of PO. It concludes that PO is a multidimensional construct, consisting of the following dimensions: design and documentation of business processes; management commitment towards PO; the process owner role; process performance measurement; a corporate culture in line with the process approach; application of continuous process improvement methodologies; and process-oriented organizational structure.

Research limitations/implications – The sample used in this work only included Austrian firms operating in manufacturing industry. Generalizability of the findings to other industries or other countries is open to scrutiny.

Originality/value – Several studies into process management use proxy variables as an indicator for PO (e.g. ISO 9000 certification). Other studies treat the construct as a unidimensional measure. However, unidimensional measures meet with increasing criticism, as they turn out to be insufficient to capture the richness of such a complex construct. This paper attempts to rectify the content deficiencies in the literature on the construct by specifying its theoretical domain and dimensionalization.

Keywords Process management, Austria, Measurement, Manufacturing industry

Paper type Literature review

1. Introduction

Organizational process alignment, which is also often referred to as (business) process orientation (PO), can be interpreted as the organizational effort required making business processes the platform for organizational structure and strategic planning (Reijers, 2006; Sabherwal *et al.*, 2001; Suter, 2009). A process-oriented organization is also often referred to as “horizontal organization” (Ostroff, 1999), “process-centered organization” (Hammer, 1996), “process enterprise” (Hammer and Stanton, 1999), “process-focused organization” (Gardner, 2004) or simply “process organization” (Osterloh and Frost, 2006; Gaitanides, 2007). A process-oriented organization comprehensively applies the concept of business



Business Process Management
Journal
Vol. 17 No. 2, 2011
pp. 267-283

© Emerald Group Publishing Limited
1463-7154
DOI 10.1108/14637151111122347

The authors would like to thank Professor Reinhard Haberfellner from Graz University of Technology for his valuable comments.

process management. A firm which adopted the process view of its organization, regardless of whether it has already run through business process reengineering and/or process improvement projects or not, is concerned with the management of its business processes (Armistead and Machin, 1998). While there are factors such as strategy, technology, people, etc. which impact organizational performance (Thorp, 2003), the capability of managing and improving the organization's business processes represents an important determinant of firm performance as well. Many companies started to realize the benefits associated with implementing business PO (Kumar *et al.*, 2008). Several authors and empirical studies refer to a positive impact of process-oriented organizational design on firm performance (Kohlbacher, 2010). However, a lot of the existing literature of the process-oriented organization aimed at educating managers on implementing process management practices and lacks research or an empirical focus (Benner and Tushman, 2003; Sussan and Johnson, 2003). Despite the growth in the business process management literature, certain important gaps still remain. Specifically, the definition of the PO construct and its antecedents is sometime ambiguous. Clearly, the intangible nature of the construct represents a hurdle to its conceptualization, but the problems outlined above are mainly due to the fact that the literature adopts the concept without questioning its scope. Many studies into process management use proxy variables as an indicator for PO (e.g. ISO 9000 certification). Other studies treat the construct as a unidimensional measure. However, unidimensional measures meet with increasing criticism, as they turn out to be insufficient to capture the richness of such a complex construct. This study attempts to rectify the content deficiencies in the literature on the construct by specifying its theoretical domain and dimensionalization.

The paper will begin by developing the theoretical dimensions of PO (from now on called a priori model). The study then applies a factor analysis using data from a sample of 152 Austrian manufacturing firms. The result of the factor analysis is the revised model, consisting of empirical unidimensional factors, each measuring a certain aspect of PO. The paper concludes with a discussion of the results, implications, and issues for further research.

2. Developing the a priori model

In order to define or specify a construct domain, one has to represent the construct from the literature to provide the concept with theoretical meaning, and one has to specify the dimensions that represent the construct (Churchill, 1979; DeVellis, 1991).

The approach of PO emphasizes processes as opposed to hierarchies (McCormack and Johnson, 2001). PO means focusing on business processes ranging from customer to customer instead of placing emphasis on functional structures (Reijers, 2006). It capitalizes on improving an organization's efficiency through high-level coordination of an organization's activities in a rationalized system of end-to-end processes (Benner and Tushman, 2002). The process management philosophy is a comprehensive problem-solving heuristic that is process oriented, customer focused, fact based, and participative throughout a firm (Winter, 1994). Business process management incorporates the discovery, design, deployment, execution, interaction, control, analysis, and optimization of business processes (Smith and Fingar, 2003). These definitions have proved cardinal to the conceptualization of the PO construct, to such an extent that very few studies have revised or expanded this definition.

Shenkar *et al.* (1995) stipulate that the definition of a construct domain must fulfill two requirements: the construct must be testable, in other words, one must be able to operationalize its dimensions; and the construct must also be global, that is, incorporate the dimensions that different classifications propose. Following the line of authors who did extensive research in the field of PO (Fischermanns, 2006; Hammer, 2007a; McCormack and Johnson, 2001; Reijers, 2006; Vera and Kuntz, 2007) and in accordance with the authors' interpretation of the construct, nine different dimensions cover the domain of PO which will be discussed in the next section. Each dimension is operationalized by several items by using Likert-type self-evaluation scales. In total, this study uses 37 indicators for measuring the PO construct. To enhance the construct validity, the resulting model was discussed and approved by three chief process officers from different organizations operating in different industries. The nine dimensions of the a priori model are:

- (1) process design and documentation;
- (2) management commitment;
- (3) process owner;
- (4) process performance measurement;
- (5) corporate culture in line with the process approach;
- (6) organizational structure in line with the process approach;
- (7) people and expertise;
- (8) process-oriented HR systems; and
- (9) coordination and integration of process projects.

The indicators of each dimension are described in the following.

2.1 Process design and documentation

Business processes present a difficult challenge in identification and analysis since they are often unknown quantities, have no names, and are not visualized in organizational charts (Kiraka and Manning, 2005). A prerequisite for managing an organization based on its processes is to know which business processes are performed within the organization and how they are related to each other. A precise definition of the company's business processes is the starting point for process management (Hinterhuber, 1995). This dimension measures to which degree the firm's processes are explicitly designed and documented and includes the following indicators:

- *Existence of a complete and uniform enterprise process model (MACROM).* The enterprise process model, which is also sometimes referred to as "macro enterprise process map" (Gardner, 2004), gives an overview of the organization's business processes. This indicator is adopted from the models of Fischermanns (2006) and Hammer (2007a) and captures whether the firm has developed a complete and uniform enterprise process model illustrating the business processes of the organization.
- *Documentation of processes (DOC).* Business processes need to be specified of how they are to be executed (Hammer, 2007a). This indicator is adopted from the models by Fischermanns (2006), Hammer (2007a), McCormack and

Johnson (2001), Reijers (2006) and Willaert *et al.* (2007) and measures whether the firm's processes are documented in a sufficient detailed way.

- *Use and update of process documentation (DOCUSE)*. Without a timely update of the documentation after a change of the process design, people will soon discontinue to use the documentation, making the process documentation largely useless. This measure is adopted from the models by Reijers (2006) and Willaert *et al.* (2007) and measures whether process documentation is always timely updated after process design has changed.
- *Definition of inputs and outputs for each process (INPOUTP)*. The specification of a business process also needs to include a definition of the process' inputs and outputs (Schantin, 2004). This measure is adopted from the models by Hammer (2007a) and McCormack and Johnson (2001), and captures whether the inputs and outputs of the firm's processes are clearly defined.
- *Definition of customers and suppliers for each process (CUSTSUPP)*. A main characteristic of a process is that it is initiated by, and must provide results to, a customer (Davenport and Short, 1990; Childe *et al.*, 1994; Schantin, 2004; Suter, 2004). This measure is adopted from the models by Hammer (2007a) and McCormack and Johnson (2001) and measures whether the internal/external customers as well as the internal/external suppliers of the organization's processes are clearly defined.
- *Segmentation of business processes (SEGMENT)*. The idea of business process segmentation was originally developed by Schantin (2004) and Suter (2004). Also, other authors refer to this idea. For instance, Osterloh and Frost (2006) call it "Triage" and Gaitanides (2007) calls it "process variant". Segmentation of a business process refers to the idea of creating process variants of a business process which faces heterogeneous market and/or customer requirements. The objective is that every process variant created can then handle homogeneous requirements. This indicator captures whether the firm's process design uses the idea of process segmentation, i.e. whether the organization has built several process variants for business processes that face heterogeneous requirements. Although this indicator is not used by any other model or study operationalizing PO, it is included in this study.

2.2 Management commitment

In a process-oriented organization, management needs to support the process program. Without the support of senior executives, the process idea cannot unfold to its full potential. There is a high risk for process management to fail if the senior executives do not undertake necessary leadership roles and do not promote process-oriented thinking (Hinterhuber, 1995). Process-oriented initiatives are less likely to secure benefits unless managers come to a consensus and an understanding of such initiatives (Edwards *et al.*, 2000). This dimension captures to what extent management supports the process program. Characteristical items for investigation are:

- *Senior executive team's percipience of process management (MGMT_AW)*. PO must be a long-term commitment rather than a quick fix. This indicator is derived from the model by Hammer (2007a) and captures whether management perceives process management not as a single project, but as a way of managing the business.

- *Existence of a senior executive who has taken leadership of and responsibility for the process program (CPO)*. In an ideal case, the organization has established a so-called chief process officer (CPO) who deeply understands the concept of the process approach and who is centrally responsible for the advancement of the enterprise-wide business process management (Schmelzer and Sesselmann, 2006). This indicator is derived from the model by Hammer (2007a) and measures whether there is at least one senior executive who has taken leadership of, and responsibility for, the process program.
- *Active engagement of the senior executive team in the process program (MGMT_BEH)*. Participation of the senior executive team in, e.g. setting process performance goals or deciding between several process design variants is included as an indicator of management commitment towards the process program. This indicator is derived from the models by Hammer (2007a) and Willaert *et al.* (2007) and captures whether the senior executive team is actively engaged in the process program.

2.3 Process owner

The existence of process owners is the most visible difference between a process enterprise and a traditional organization (Hammer and Stanton, 1999). This dimension measures the extent to which the role of the process owner is implemented in the firm:

- *Existence of process owners (PO_EX)*. A business process needs to have a process owner having end-to-end responsibility of the process (Suter, 2009). This indicator is adopted from the models by Fischermanns (2006), Hammer (2007a), Reijers (2006) and Willaert *et al.* (2007) and measures whether the role of the process owner is established in the organization and whether a process owner has been assigned to each business process.
- *Experience of process owners (PO_MNGR)*. A process owner needs to have leadership experience (Schmelzer and Sesselmann, 2006). This measure is adopted from the model by Hammer (2007a) and captures whether process owners are experienced leaders/managers.
- *Power of the process owner in order to be able to act for the process interests (PO_ID)*. Process owners have to have the authority to take all measures necessary to coordinate and improve the business process (Hinterhuber, 1995). This measure is adopted from the model by Hammer (2007a) and captures whether process owners are members of the enterprise's senior-most decision-making body.
- *Are process owners responsible for continuous improvement of their processes and do they perform this task proactively? (PO_ACT)*. An important task of a process owner is the continuous improvement and optimization of the process the owner is responsible for (Schmelzer and Sesselmann, 2006). This measure is adopted from the model by Hammer (2007a) and captures whether process owners are responsible for continuous improvement of their processes and whether they perform this task proactively.
- *Process owner's responsibility for budget (PO_BUD)*. In an ideal process-oriented organization, budgeting is done by process instead of by department (Hammer and Stanton, 1999). This measure is adopted from the model by Hammer (2007a) and captures whether process owners bear budget responsibility.

- *Process owner's influence over personnel assignments (PO_STAFF)*. Process performance logically depends on the workers executing the actual work within the process. If process owners have influence over personnel assignments, they thereby also have more influence on the performance of the process. This measure is adopted from the model by Hammer (2007a) and captures whether process owners have strong influence over personnel assignments.

2.4 Process performance measurement

By focusing measurement on processes rather than functions, alignment and common focus across separate organizational units can be achieved (Hammer, 2007b). Implementing measures and taking corrective actions are operating precepts of process management (Melan, 1989), since a business process can only be mastered if it can be measured (Hinterhuber, 1995). This dimension measures to which degree measurement of process performance is carried out in the organization:

- *Existence of process performance indicators for business processes (PPM_DEF)*. Process performance indicators are metrics capturing the performance of a business process. This indicator is derived from the models by Fischermanns (2006), Hammer (2007a), McCormack and Johnson (2001), Reijers (2006) and Willaert *et al.* (2007) and captures whether performance indicators are defined for business processes.
- *Derivation of process performance indicators from enterprise goals (PPM_DRV)*. Process performance indicators have to be derived from the process objectives which themselves have to be derived from business objectives. This is crucial because business processes have to be aligned with business strategy (Ndede-Amadi, 2004). This indicator is derived from the model by Hammer (2007a) and captures whether process performance indicators are derived from enterprise goals and/or from (internal) customer requirements.
- *Continuous collection of process performance data (PPM_COLL)*. Process performance measurement only makes sense if performance indicators are calculated from process performance data which are collected continuously. This indicator is derived from the models by Fischermanns (2006), Hammer (2007a), McCormack and Johnson (2001), Reijers (2006) and Willaert *et al.* (2007) and captures whether process performance data are collected continuously.
- *Are actions actually initiated by performance indicators? (PPM_ACT)*. Measuring process performance without reacting on poor performance stresses resources, but does not lead to any improvements. This indicator is derived from the models by Fischermanns (2006), Hammer (2007a) and Reijers (2006) and captures whether improvement actions are actually initiated if poor process performance is encountered.
- *Presentation of metrics to process workers (PPM PRES)*. Process workers who know the performance of the business process are able to timely react on bad performance and are therefore considered as really empowered. This indicator is derived from the model by Hammer (2007a) and captures whether process metrics are presented periodically to process performers (e.g. awareness and motivation).
- *Use of metrics for process benchmarking (BENCHM)*. Process benchmarking uses business processes as comparison units and aims to identify best operating

practices (Delpachitra and Beal, 2002). This measure is adopted from the model by Hammer (2007a) and captures whether process benchmarking is carried out in the organization.

- *Use of activity-based costing (ABCOST)*. Activity-based costing captures costs horizontally in line with business processes (Hinterhuber, 1995). This measure is adopted from the models by Hammer (2007a), Reijers (2006) and Vera and Kuntz (2007) and captures whether activity-based costing is comprehensively applied in the organization.

2.5 Corporate culture in line with the process approach

PO is also a matter of enterprise culture. The real problems when implementing the PO approach are of a cultural nature (Hinterhuber, 1995). The cultural fit is an important issue since people and processes must combine to produce output (Armistead and Machin, 1997). Only a culture based on teamwork, willingness to change, customer orientation, personal accountability, and a cooperative leadership style goes hand in hand with the process approach (Hammer, 2007a). This dimension measures whether the firm exhibits a culture in line with the process approach or not:

- *Existence of inter-departmental teamwork (TEAMW)*. A process-oriented organization needs a culture which values teamwork, since business processes cutting across functions must be operated by people in a team (Hammer, 2007a). This indicator was derived from the models by Hammer (2007a), McCormack and Johnson (2001), Vera and Kuntz (2007) and Willaert *et al.* (2007) and captures whether teamwork (also between different departments) can be taken for granted in the organization.
- *Customer-focused attitude of employees (CUST_FOC)*. It is often argued that organizations emphasizing functions and hierarchy are failing to focus on the customer (Hinterhuber, 1995; Schmelzer and Sesselmann, 2006; Gulati, 2007). By contrast, in a process-oriented organization, each business process has a clearly defined customer who receives the result of the process (Schantin, 2004). This indicator is derived from the models by Hammer (2007a), Reijers (2006) and Willaert *et al.* (2007) and captures whether the organization's employees understand that the purpose of their work is to fulfill the needs of the internal/external customers.
- *Employees' accountability for enterprise results (CHARGE)*. Only organizations whose culture values personal accountability will find it possible to move forward with their degree of PO (Hammer, 2007a). This indicator is derived from the model by Hammer (2007a) and captures whether the organization's employees feel accountable for enterprise results.
- *Employees' attitude towards change (ATT_CHNG)*. The lack of a change-supportive culture is often blamed when process improvement actions fail (Tenner and DeToro, 2000). As business conditions change, process designs need to evolve, and it is the task of process owners to guide that evolution (Hammer and Stanton, 1999). A remarkable part of a company's change capability is based on its employees (Nyhuis *et al.*, 2008). Taken from the model by Hammer (2007a), this indicator captures whether changes of how work is performed is accepted by the employees in a sluggish manner (this item was recoded after data collection so that a high score reflects a positive attitude towards change).

- *Use of process language (LANGUAGE)*. PO is a construct which becomes “real” by communication and interaction, i.e. the construct becomes real if it is communicated by means of a language. By communicating about business processes and their design, PO becomes a reality (Gaitanides, 2007). This indicator is derived from the models by McCormack and Johnson (2001) and Reijers (2006) and captures whether employees on all levels of the organization are speaking about business processes, customers, teams, process performance indicators, etc.
- *Existence of an open and collaborative leadership style (MGMT_ST)*. The need to empower staff is often mentioned by literature on team and process-based organizations (Armistead and Rowland, 1996b). Traditional management styles have no place in a process enterprise. Managers cannot command and control but they have to negotiate and collaborate (Hammer and Stanton, 1999). This indicator is derived from the model by Hammer (2007a) and captures whether the management’s leadership style is based on hierarchical “command and control” (this item was recoded after data collection so that a high rating reflects a collaborative leadership style to a greater extent).

2.6 Organizational structure in line with the process approach

A process-oriented organization has adapted its structure to the process view, following the principle “structure follows process” (Gaitanides, 2007). This dimension measures the extent to which organizational structure fits process design:

- *Are the organization’s business processes reflected by the organizational structure? (STRUCT1)*. This indicator was derived from the model by Reijers (2006) and captures whether the organizational structure can be best illustrated as a collection of business processes.
- *Is the organizational structure derived from processes? (STRUCT2)*. Several authors stress that business processes should determine the structure of an organization (Schantin, 2004; Suter, 2004; Osterloh and Frost, 2006; Gaitanides, 2007). This indicator was derived from the model by Willaert *et al.* (2007) and captures whether the organizational structure is derived from its business processes.

2.7 People and expertise

This dimension maps certain knowledge, expertise, and skills of the people working in a process-oriented organization:

- *Do process workers have detailed knowledge of how their process is executed? (KNOWL)*. Process performers must have appropriate knowledge, otherwise they will not be able to implement the process design (Hammer, 2007a). This indicator is derived from the model by Hammer (2007a) and captures whether the organization’s employees can describe the design of the process they work for. Furthermore, the indicator captures whether the employees know how their work affects subsequent work, customers, and process performance.
- *Are process workers skilled in decision, process improvement, and problem solving techniques? (SKILLS)*. This indicator is derived from the models by Hammer (2007a) and Willaert *et al.* (2007) and captures whether the organization’s employees are skilled in problem solving, process improvement, and decision techniques.

- *Existence of a process redesign, project management, and change management experts cadre (EXPERTS)*. According to Hammer (2007a), a process organization must have some people with skills in, and knowledge of, process redesign. This indicator is derived from the model of Hammer (2007a) and captures whether an expert cadre is available in the organization including change management, process redesign, and project management experts.
- *Use of process improvement methodologies (METHODLG)*. Improving the performance of business processes requires the knowledge about, and usage of, process improvement methodologies. This indicator was taken from the model by Hammer (2007a) and captures whether the organization makes use of methodologies for continuous process improvement, like KAIZEN, Six Sigma, etc.

2.8 Process-oriented HR systems

This dimension captures whether process-oriented HR systems are in place. In a process-oriented enterprise, the process design should drive job descriptions. Also, incentive systems should be implemented which emphasize the process' needs:

- *Are role definitions and job descriptions driven by processes? (JOB_SPEC)*. This indicator was derived from the model by Hammer (2007a) and captures whether process' design drive role definitions, job descriptions, and competency profiles.
- *Existence of incentive systems which emphasize the process' needs (INC_SYS)*. Traditional vertical management systems pull people in one direction, whereas they should work for the interest of horizontal processes. Therefore, management systems also need to emphasize the process' needs; otherwise conflict and confusion ensue, reducing performance (Hammer and Stanton, 1999). This indicator was taken from the models by Hammer (2007a), Vera and Kuntz (2007) and Willaert *et al.* (2007) and captures whether the organization has implemented reward systems (incentive systems) that emphasize the needs of business processes.

2.9 Coordination and integration of process projects

Finally, this dimension captures the existence of an instance coordinating and integrating process projects, which is also often referred to as business process management group (Harmon, 2007). This dimension is measured by one indicator (PROJ_INT) which was derived from the models by Fischermanns (2006), Hammer (2007a) and Willaert *et al.* (2007) and captures whether the organization has implemented an instance coordinating and integrating projects dealing with business process management.

3. Validating the PO measurement model

3.1 Data collection

A questionnaire was designed based on the model introduced. For all indicators, a statement was formulated. Each statement had to be rated by the respondents using a six-point Likert scale ranging from full disagreement to full agreement. The questionnaire was tested and improved by conducting cognitive interviews and a pilot study.

The population of this study is defined as Austrian corporations operating in metal and machinery industry with at least 50 employees. For practical reasons, the metal and

machinery industry was chosen since these industries include a sufficient large number of organizations in Austria. Firms were selected randomly and telephone interviews were used for data collection. All telephone interviews were personally conducted by the researcher. For every firm, one executive (CEO, CIO or quality manager) was interviewed. This is a clear difference to studies which are using a mail survey method for data collection. Data quality of this study is expected to be high since respondents are personally identified and interviewed assuring that the interviewee has the knowledge to truthfully answer the questions. A total of 152 organizations were interviewed.

3.2 Factor analysis

Since the concept of PO is a multidimensional construct, one has to ensure that the construct's underlying dimensions are unidimensional, reliable, and valid. Unidimensionality is assessed by exploratory factor analysis. In order to determine the appropriateness of a factor analysis application, a Bartlett test of sphericity is performed, as recommended by Hair *et al.* (2006). The test finds that the correlation matrix has significant correlations, justifying the factor analysis. In addition, as recommended by Hair *et al.* (2006), the appropriateness of factor analysis is assessed by the Kaiser-Meyer-Olkin's measure of sampling adequacy (MSA). The overall MSA value accounts for 0.912, which highly justifies the application of a factor analysis. As described by Hair *et al.* (2006), MSA should also be applied for each individual variable entering the factor analysis. The MSA values of the 37 PO items are all over the recommended value of 0.5 (none of the values are below 0.775), also ensuring the appropriateness of a factor analysis application. The initial principal components analysis, carried out with varimax rotation, yielded to seven factors with eigenvalues greater than the traditional cutoff of 1.0. These seven factors retained 69.31 percent of the total variance in the data. However, four items have been identified which loaded not more than 0.50 on any factor: existence of an open and collaborative leadership style (MGMT_ST), process workers' skills in decision, process improvement and problem solving techniques (SKILLS), conformance of the organization's business processes with its organizational structure (STRUCT1), and role definitions and job descriptions derived from process design (JOB_SPEC). These items have been dropped from the analysis, since only factor loadings ± 0.5 or greater are considered practically significant (Hair *et al.*, 2006). As four items have been removed from the analysis, a new factor analysis for the remaining 33 items was performed, which yielded to the expected seven factors retaining 72.52 percent of the total variance of the 33 items. Construct validity is supported by the fact that none of the items loaded greater than 0.50 on more than one factor. Adequate construct reliability of the factors was assessed by using Cronbach's alpha. The final factor solution consists of the following dimensions.

Process performance measurement ($\alpha = 0.937$), including the items PPM_PRES (presentation of process performance metrics to workers), PPM_COLL (continuous collection of process performance data), PPM_ACT (action initiation by process performance indicators), PPM_DRV (derivation of process performance indicators from enterprise goals), PPM_DEF (existence of process performance indicators for processes), BENCHM (use of process benchmarking), ABCOST (use of activity-based costing), and INC_SYS (existence of incentive systems emphasizing the process' objectives). This revised dimension slightly differs from the a priori dimension as the INC_SYS item has shifted to this dimension. This shift is theoretically founded since incentive

systems based on the process' objectives can only be realized if process performance measurement is implemented (the incentive systems require process performance measurement in place). This is also clearly addressed by Armistead and Rowland (1996a) stating that process performance measurement plays a role in setting process targets for improvement, and these process targets are linked to reward.

Process owner ($\alpha = 0.943$), including the items PO_ID (process owners' power to be able to act for the process interest), PO_MNGR (are process owners experienced leaders/managers?), PO_STAFF (process owners' influence over personnel assignments), PO_ACT (process owners' responsibility for continuous improvement of their processes and proactive execution of this task), PO_BUD (process owner's responsibility for budget), and PO_EX (existence of process owners). This dimension completely complies with the a priori dimension.

Design and documentation of business processes ($\alpha = 0.885$), including the items DOC (design and documentation of processes), CUSTSUPP (definition of processes' customers and suppliers), INPOUTP (definition of processes' inputs and outputs), DOCUSE (update of process documentation), MACROM (existence of macro process model), and SEGMENT (existence of process variants if applicable). This dimension completely complies with the a priori dimension.

Management commitment ($\alpha = 0.882$), including the items MGMT_BEH (active engagement of senior executive team in the process program), CPO (existence of a senior executive who has taken leadership and responsibility of the process program), PROJ_INT (existence of an instance coordinating and integrating process projects), and MGMT_AW (management's perception of process management as a way of managing the business). This revised dimension slightly differs from the a priori dimension since PROJ_INT is included in the revised dimension. A possible interpretation for this inclusion is that an active engagement of the senior executive team in the process program actually could be done in the form of integrating and coordinating process programs (at least in smaller organizations).

Culture and people in line with the process approach ($\alpha = 0.814$), including the items CUST_FOC (customer-focused attitude of employees), CHARGE (employees' accountability for firm results), TEAMW (existence of teamwork), KNOWL (workers' knowledge towards how processes are executed), and LANGUAGE (use of process language). The KNOWL item (knowledge of process workers about how the process is executed and about how process results influence subsequent work and/or customers) has shifted to this dimension. By knowing how processes are executed, the process approach is lived by the employees and represents therefore an aspect of a corporate culture in line with the process approach.

Continuous improvement methodologies ($\alpha = 0.631$), including METHODLG (use of process improvement methodologies) and EXPERTS (existence of process redesign, project management and change management experts). This is a completely new dimension with a sound theoretical basis: continuous process improvement methodologies can be deployed only if certain knowledge about redesign, project management and change management is present in the company.

Process-oriented organizational structure ($\alpha = 0.523$), including ATT_CHNG (employees' attitude towards change) and STRUCT2 (derivation of the organizational structure from process design). At first glance hard to interpret, this dimension represents the ability of the organization to align the structure with the processes which

is accompanied by the ability to implement organizational changes without major resistance of the employees. As noted by Themistocleous and Corbitt (2006), improving existing processes often requires changes in the organizational structure; and according to Nyhuis *et al.* (2008), a remarkable part of a company's change capability is based on its employees. The objective factor analysis outputs a factor which apparently is representing a firm's capability of aligning its structure to its business processes, which is enabled by the employees' positive attitude towards change.

As reported above, adequate construct reliability of the dimensions was assessed by using Cronbach's alpha. The coefficients were all above the minimum acceptance level of 0.6 (Hair *et al.*, 2006), except the coefficient for the last factor (process-oriented organizational structure), indicating that the factor has less than desirable internal consistency. However, in early stages of research, according to Nunnally (1978), one can work with instruments that have only modest reliability (and the construct of process-oriented organizational structure definitely is subject of early stage research). Therefore, the dimension is retained in the study and interpretation regarding this variable needs to be made with caution. The results of the factor analysis, i.e. the revised model, are summarized in Table I. The table shows the explained variance, Cronbach's alpha values, as well as the items and item loadings of each dimension.

4. Conclusion

This study advances the understanding of the PO construct by developing a measurement model incorporating the multidimensional nature of the construct. The study suggests that PO is a construct consisting of seven dimensions:

- (1) design and documentation of business processes;
- (2) management commitment towards PO;
- (3) the process owner role;
- (4) process performance measurement;
- (5) a corporate culture in line with the process approach;
- (6) application of continuous process improvement methodologies; and
- (7) process-oriented organizational structure.

The model is grounded in a thorough review of the literature and its dimensions exhibit satisfactory unidimensionality, reliability, and validity. At present, many managers are looking for ways to make their organization more process oriented. Our findings indicate that business process management involves many different aspects, ranging from process performance measurement to process-oriented organizational structure. The developed model has potential use for an organization to review its internal progress of process-oriented organizational design and has potential use for process assessment, either as an alternative measurement or supplementary measurement of process capability and organizational maturity. Including as many as 33 items in order to assess process-oriented organizational design might be a problem from a practical point of view. However, in order to get a complete assessment, one should use the whole range of items. In summary, we advance the growing body of literature on process management by contributing to a greater clarity and better understanding of the PO construct and its dimensions.

Dimension	Variance explained (%)	Cronbach alpha	Item	Factor loading
Process performance measurement	42.34	0.937	Presentation of process performance metrics to workers (PPM_PRES)	0.842
			Continuous collection of process performance data (PPM_COLL)	0.816
			Action initiation by process performance indicators (PPM_ACT)	0.780
			Derivation of process performance indicators from enterprise goals (PPM_DRV)	0.764
			Existence of process performance indicators for processes (PPM_DEF)	0.750
			Use of process benchmarking (BENCHM)	0.741
			Use of activity-based costing (ABCOST)	0.601
			Existence of incentive systems emphasizing the process' objectives (INC_SYS)	0.586
			Process owners' power to be able to act for the process interest (PO_ID)	0.865
			Are process owners experienced leaders/managers? (PO_MNGR)	0.809
Process owner	7.94	0.943	Process owners' influence over personnel assignments (PO_STAFF)	0.800
			Process owners' responsibility for continuous improvement of their processes and proactive execution of this task (PO_ACT)	0.744
			Process owner's responsibility for budget (PO_BUD)	0.736
			Existence of process owners (PO_EX)	0.678
			Design and documentation of processes (DOC)	0.782
			Definition of processes' customers and suppliers (CUSTSUPP)	0.731
			Definition of processes' inputs and outputs (INPOUTP)	0.714
			Update of process documentation (DOCUSE)	0.687
				(continued)
Design and documentation of business processes	5.64	0.885		

Table I.
PO dimensions, Cronbach
alpha values, and factor
loadings of the revised
model

Table I.

Dimension	Variance explained (%)	Cronbach alpha	Item	Factor loading
Management commitment	5.16	0.882	Existence of macro process model (MACROM)	0.642
			Existence of process variants if applicable (SEGMENT)	0.588
			Active engagement of senior executive team in the process program (MGMT_BEH)	0.797
			Existence of a senior executive who has taken leadership and responsibility of the process program (CPO)	0.786
Culture and people in line with the process approach	4.61	0.814	Existence of an instance coordinating and integrating process projects (PROJ_INT)	0.715
			Management's perception of process management as a way of managing the business (MGMT_AW)	0.707
			Customer-focused attitude of employees (CUST_FOC)	0.802
			Employees' accountability for firm results (CHARGE)	0.696
Continuous improvement methodologies	3.58	0.631	Existence of teamwork (TEAMW)	0.686
			Workers' knowledge towards how processes are executed (KNOWL)	0.656
			Use of process language (LANGUAGE)	0.555
			Use of process improvement methodologies (METHODLG)	0.740
Process-oriented organizational structure	3.25	0.523	Existence of process redesign, project management, and change management experts (EXPERTS)	0.535
			Employees' attitude towards change (ATT_CHNG)	0.788
			Derivation of the organizational structure from process design (STRUCT2)	0.587

Notes: The displayed “explained variance” is after varimax rotation; cumulative variance explained is 72.52 per cent

As with any empirical study, this work has obvious limitations that should be recognized. First, the sample in this work only included Austrian manufacturing companies. Although one should expect similar findings in similar industries, generalizability of the findings to other industries or other countries is open to scrutiny. However, no specific reason points to the fact that nationality might bias the results in a predictable direction. Second, only one interview per firm was conducted. Interviewing several managers per firm would have led to even higher data quality. A third important limitation of this work is the small sample size. Such a small number of cases is not appropriate for a solid validation of the measurement model. Fourth, the model's dimension measuring process-oriented organizational structure exhibits non-satisfactory internal consistency. Much work remains to be done to improve the understanding and conceptualization of the process-oriented organizational structure dimension.

There are a number of avenues for further research. First, this study could be applied to other industries (particularly interesting would be service industries) and/or other countries. Second, other methodologies, such as a structural equation model, could be used to assess the underlying dimensions of the PO construct. However, this would require more data in order to obtain reasonable results. Third, the introduction of size as a moderating variable in the suggested measurement model is another interesting area for future research. The inclusion of this variable allows us to test the equivalence of the proposed measurement model across groups of firms based on size. Fourth, we treated the PO construct and its dimensions as reflective measures. Future studies should build a measurement model which considers the construct and the dimensions as formative measures and compare the results to this paper. Finally, future studies could use the measurement model to investigate the effects of PO dimensions on organizational performance. This empirical research is needed in order to determine whether the outcome effects of different specific aspects of PO are linear or additive or if they also involve interactions and diminishing returns. Such studies are – to date – remarkably limited.

References

- Armistead, C. and Machin, S. (1997), "Implications of business process management for operations management", *International Journal of Operations & Production Management*, Vol. 17 No. 9, pp. 886-98.
- Armistead, C. and Machin, S. (1998), "Business process management: implications for productivity in multi-stage service networks", *International Journal of Service Industry Management*, Vol. 9 No. 4, pp. 323-36.
- Armistead, C. and Rowland, P. (1996a), "Managing by business processes", in Armistead, C. and Rowland, P. (Eds), *Managing Business Processes: BPR and Beyond*, Wiley, Chichester, pp. 39-59.
- Armistead, C. and Rowland, P. (1996b), "The role of people in processes", in Armistead, C. and Rowland, P. (Eds), *Managing Business Processes: BPR and Beyond*, Wiley, Chichester, pp. 61-71.
- Benner, M.J. and Tushman, M. (2002), "Process management and technological innovation: a longitudinal study of the photography and paint industries", *Administrative Science Quarterly*, Vol. 47, pp. 676-706.
- Benner, M.J. and Tushman, M.L. (2003), "Exploitation, exploration, and process management: the productivity dilemma revisited", *The Academy of Management Review*, Vol. 28 No. 2, pp. 238-56.

- Childe, S.J., Maull, R.S. and Bennett, J. (1994), "Frameworks for understanding business process re-engineering", *International Journal for Operations & Production Management*, Vol. 14 No. 12, pp. 22-34.
- Churchill, G.A. (1979), "A paradigm for developing better measures of marketing constructs", *Journal of Marketing Research*, Vol. 16 No. 1, pp. 64-73.
- Davenport, T.H. and Short, J.E. (1990), "The new industrial engineering: information technology and business process redesign", *Sloan Management Review*, Vol. 31 No. 4, pp. 11-27.
- Delpachitra, S. and Beal, D. (2002), "Process benchmarking: an application to lending products", *Benchmarking: An International Journal*, Vol. 9 No. 4, pp. 409-20.
- DeVellis, R.F. (1991), *Scale Development: Theory and Applications*, Sage, Newbury Park, CA.
- Edwards, C., Braganza, A. and Lambert, R. (2000), "Understanding and managing process initiatives: a framework for developing consensus", *Knowledge and Process Management*, Vol. 7 No. 1, pp. 29-36.
- Fischermanns, G. (2006), *Praxishandbuch Prozessmanagement*, Dr. Götz Schmidt, Gießen.
- Gaitanides, M. (2007), *Prozessorganisation*, Vahlen, Munich.
- Gardner, R. (2004), *The Process-Focused Organization*, Quality Press, Milwaukee, WI.
- Gulati, R. (2007), "Abschied vom Silodenken", *Harvard Business Manager*, Vol. 12, pp. 90-106.
- Hair, J.F., Black, W.C., Babin, B.J., Anderson, R.E. and Tatham, R.L. (2006), *Multivariate Data Analysis*, Pearson Prentice-Hall, Upper Saddle River, NJ.
- Hammer, M. (1996), *Beyond Reengineering*, Harper Collins, London.
- Hammer, M. (2007a), "The process audit", *Harvard Business Review*, Vol. 85 No. 4, pp. 111-23.
- Hammer, M. (2007b), "The 7 deadly sins of performance measurement", *MIT Sloan Management Review*, Vol. 48, pp. 19-28.
- Hammer, M. and Stanton, S. (1999), "How process enterprises really work", *Harvard Business Review*, Vol. 77 No. 6, pp. 108-18.
- Harmon, P. (2007), *Business Process Change*, Morgan Kaufmann, Burlington, MA.
- Hinterhuber, H.H. (1995), "Business process management: the European approach", *Business Change & Re-engineering*, Vol. 2 No. 4, pp. 63-73.
- Kiraka, R.N. and Manning, K. (2005), "Managing organizations through a process-based perspective: its challenges and benefits", *Knowledge and Process Management*, Vol. 12 No. 4, pp. 288-98.
- Kohlbacher, M. (2010), "The effects of process orientation: a literature review", *Business Process Management Journal*, Vol. 16 No. 1, pp. 135-52.
- Kumar, U., Lavassani, K.M., Kumar, V. and Movahedi, B. (2008), "Measurement of business process orientation in transitional organizations: an empirical study", in Abramowicz, W. and Fensel, D. (Eds), *Business Information Systems*, Springer, Berlin, pp. 357-68.
- McCormack, K.P. and Johnson, W.C. (2001), *Business Process Orientation: Gaining the E-Business Competitive Advantage*, St Lucie Press, Boca Raton, FL.
- Melan, E.H. (1989), "Process management: a unifying framework for improvement", *National Productivity Review*, Vol. 8 No. 4, pp. 395-406.
- Ndede-Amadi, A.A. (2004), "What strategic alignment, process redesign, enterprise resource planning, and e-commerce have in common: enterprise-wide computing", *Business Process Management Journal*, Vol. 10 No. 2, pp. 184-99.
- Nunnally, J.C. (1978), *Psychometric Theory*, McGraw-Hill, New York, NY.

-
- Nyhuis, P., Heinen, T. and Steinke, T. (2008), "Was die Wandlungsfähigkeit von Fabriken bestimmt", *IO New Management*, No. 10, pp. 31-5.
- Osterloh, M. and Frost, J. (2006), *Prozessmanagement als Kernkompetenz*, Gabler, Wiesbaden.
- Ostroff, F. (1999), *The Horizontal Organization*, Oxford University Press, Oxford.
- Reijers, H.A. (2006), "Implementing BPM systems: the role of process orientation", *Business Process Management Journal*, Vol. 12 No. 4, pp. 389-409.
- Sabherwal, R., Hirschheim, R. and Goles, T. (2001), "The dynamics of alignment: insights from a punctuated equilibrium model", *Organization Science*, Vol. 12 No. 2, pp. 179-97.
- Schantin, D. (2004), *Makromodellierung von Geschäftsprozessen*, Deutscher Universitätsverlag, Wiesbaden.
- Schmelzer, H.J. and Sesselmann, W. (2006), *Geschäftsprozessmanagement in der Praxis*, Carl Hanser, Munich.
- Shenkar, O., Aranya, N. and Almor, T. (1995), "Construct dimensions in the contingency model: an analysis comparing metric and non-metric multivariate instruments", *Human Relations*, Vol. 48 No. 5, pp. 559-80.
- Smith, H. and Fingar, P. (2003), *Business Process Management: The Third Wave*, Meghan-Kiffer Press, Tampa, FL.
- Sussan, A.P. and Johnson, W.C. (2003), "Strategic capabilities of business processes: looking for competitive advantage", *Competitiveness Review*, Vol. 13 No. 2, pp. 46-52.
- Suter, A. (2004), *Die Wertschöpfungsmaschine*, Industrielle Organisation, Zurich.
- Suter, A. (2009), *Neues Wachstum*, Industrielle Organisation, Zurich.
- Tenner, A.R. and DeToro, I.J. (2000), *Process Redesign: The Implementation Guide for Managers*, Prentice-Hall, Upper Saddle River, NJ.
- Themistocleous, M. and Corbitt, G. (2006), "Is business process integration feasible?", *Journal of Enterprise Information Management*, Vol. 19 No. 4, pp. 434-49.
- Thorp, J. (2003), *The Information Paradox – Realizing the Business Benefits of Information Technology*, McGraw-Hill, Toronto, ON.
- Vera, A. and Kuntz, L. (2007), "Process-based organization design and hospital efficiency", *Health Care Management Review*, Vol. 32 No. 1, pp. 55-65.
- Willaert, P., Van den Bergh, J., Willems, J. and Deschoolmeester, D. (2007), "The process-oriented organization: a holistic view", in Dadam Alonso, P. and Rosemann, M. (Eds), *BPM 2007*, LNCS 4714, Springer, Berlin, pp. 1-15.
- Winter, S. (1994), "Organizing for continuous improvement: evolutionary theory meets the quality revolution", in Baum, J. and Singh, J. (Eds), *Evolutionary Dynamics of Organizations*, Oxford University Press, New York, NY, pp. 90-108.

Corresponding author

Markus Kohlbacher can be contacted at: markus.kohlbacher@processorientation.com